

- 1 (a) (i) scalar quantity has magnitude (allow size) B1
vector quantity has magnitude and direction B1 [2]
- (ii) 1. temperature: scalar B1 [1]
2. acceleration: vector B1 [1]
3. resistance: scalar B1 [1]
- (b) *either* triangle / parallelogram with correct shape C1
tension = 14.3 N (allow ± 0.5 N) A2 [3]
- (if $> \pm 0.5$ N but $\leq \pm 1$ N, allow 1 mark)
- or $R = 25 \cos 35^\circ$ (C1)
 $T = R \tan 35^\circ$ (C1)
 $T = 14.3$ N (A1)
- or $T = 25 \sin 35^\circ$ (C2)
 $T = 14.3$ N (A1)
- or R and T resolved vertically and horizontally (C2)
leading to $T = 14.3$ N (A1)

- 2 (a) (i) $V_x = 12.4 \cos 36^\circ (= 10.0 \text{ m s}^{-1})$